



GLOBAL ECONOMICS
GRADE: 10
LEARNING EVIDENCE 2
DECISION-MAKING UNDER UNCERTAINTY
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1ST TERM
2026–2027

Learning objective: Apply minimax regret and Expected Monetary Value (EMV) to select an alternative under uncertainty, and use sensitivity analysis to determine how a subjective probability changes the preferred alternative.

Assessment criteria:

C5: Constructs opportunity loss tables and applies the Minimax regret criterion to select the alternative that minimises the maximum opportunity loss.
 C6: Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff.
 C7: Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

Criteria assessment

Each problem is worth 2 points. Show all calculations, including the complete regret table for C5, every EMV for C6, and the threshold equation and decision ranges for C7. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. A minimum of 4 points is required to pass each criterion.

Question	C5	C6	C7
1	2	–	–
2	2	–	–
3	2	–	–
4	–	2	–
5	–	2	–
6	–	2	–
7	–	–	2
8	–	–	2
9	–	–	2
Total	6	6	6

1. Artisan Apiary Expansion

[C5-2]

An artisan apiary must choose one of three hive-expansion plans. Annual net benefits are in thousands of dollars.

Alternative	Excellent	Average	Poor
Large (A_1)	76	34	-14
Medium (A_2)	58	49	23
Small (A_3)	38	43	35

What decision would be selected using the Minimax Regret criterion? Justify your answer.

2. Regional Theater Capacity

[C5-2]

A theater operator is choosing a seating-capacity plan. Seasonal profit is in thousands of dollars; no reliable probabilities are available.

Alternative	High	Normal	Low
Add two seating sections (A_1)	108	40	-28
Add one seating section (A_2)	82	61	14
Lease a nearby hall (A_3)	59	52	34
Current seating (A_4)	34	34	34

What decision would be selected using the Minimax Regret criterion? Justify your answer.

3. Emergency Shelter Staffing [C5-2]

A relief organization must select an emergency-shelter staffing plan. Net community value is in thousands of dollars.

Alternative	Very high	High	Moderate	Low
Full (A_1)	96	54	24	-24
Flexible (A_2)	74	64	49	12
Limited (A_3)	48	51	46	38

What decision would be selected using the Minimax Regret criterion? Justify your answer.

4. Coffee Roastery Product Mix [C6-2]

A coffee roastery is selecting a product mix. Profits are in thousands of dollars.

	Strong	Stable	Weak
$P(S_j)$	0.45	0.35	0.20
Single-origin (A_1)	86	44	-12
Balanced (A_2)	64	58	22
Classic (A_3)	42	47	37

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

5. Aquarium Program Scale [C6-2]

An aquarium is deciding the scale of a temporary education program. Payoffs are in thousands of dollars.

	High	Expected	Low
$P(S_j)$	0.30	0.40	0.30
Major (A_1)	98	38	-24
Standard (A_2)	73	53	18
Compact (A_3)	52	47	34
Regular tours only (A_4)	27	27	27

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

6. Wind-Farm Maintenance Staffing [C6-2]

A wind-farm operator is choosing a maintenance staffing plan. Payoffs are in thousands of dollars.

	Surge	Strong	Steady	Slow
$P(S_j)$	0.20	0.30	0.35	0.15
Rapid hiring (A_1)	105	64	24	-32
Moderate hiring (A_2)	79	68	48	12
Current staff (A_3)	53	55	51	42

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

7. Concert Merchandise Order [C7-2]

A concert vendor must choose between a large and a small merchandise order. Payoffs are in thousands of dollars.

Alternative	High (S_1)	Low (S_2)
Large (A_1)	86	12
Small (A_2)	48	38

Let $P(S_1) = p$. State the preferred alternative for every probability value ($0 \leq p \leq 1$).

8. Meal-Kit Launch [C7-2]

A meal-kit company is comparing a bold launch with a measured launch. Payoffs are in thousands of dollars.

Alternative	High	Moderate	Low
Bold (A_1)	92	32	-8
Measured (A_2)	47	53	27

Suppose $P(S_1) = p$, $P(S_2) = \frac{p}{3}$. State the preferred alternative for every feasible probability value of p and identify the threshold probability at which the company is indifferent between the two alternatives.

9. Publishing-Platform Bid [C7-2]

A publishing firm must choose between a standard platform bid and an experimental platform bid. Payoffs are in thousands of dollars.

Alternative	Immediate	Revision	Delayed
Standard (A_1)	24	58	79
Experimental (A_2)	74	43	32

Suppose $P(S_1) = 3p$, $P(S_2) = p$. State the preferred alternative for every feasible probability value of p and identify the threshold probability at which the firm is indifferent between the two alternatives.